Similarly, the flow map of the system with Hamiltonian $H_2 = U$ is

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q}, \\ \boldsymbol{P} &= \boldsymbol{p} - h\nabla U(\boldsymbol{q}). \end{aligned}$$

The composition of these maps simplifies to

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{P}, \\ \boldsymbol{P} &= \boldsymbol{p} - h\nabla U(\boldsymbol{q}), \end{aligned}$$

which is precisely the Symplectic Euler method. Similarly, composing the same two maps in the opposite order gives the adjoint Symplectic Euler method.

*Example 2.5 (Verlet is Symplectic)* For the Symplectic Euler method $\mathscr{G}_h$ and its adjoint method $\mathscr{G}_h^\dagger$, consider the composition

$$\mathcal{K}_h := \mathscr{G}_{h/2}^\dagger \circ \mathscr{G}_{h/2}.$$

Let

$$\begin{bmatrix} \bar{\boldsymbol{q}} \\ \bar{\boldsymbol{p}} \end{bmatrix} = \mathscr{G}_{h/2} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{p} \end{bmatrix},$$

i.e.,

$$\bar{\boldsymbol{q}} = \boldsymbol{q} + (h/2)\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \qquad \bar{\boldsymbol{p}} = \boldsymbol{p} - (h/2)\nabla U(\boldsymbol{q}).$$

and set

$$\begin{bmatrix} \boldsymbol{Q} \\ \boldsymbol{P} \end{bmatrix} = \mathscr{G}_{h/2}^\dagger \begin{bmatrix} \bar{\boldsymbol{q}} \\ \bar{\boldsymbol{p}} \end{bmatrix},$$

i.e.,

$$\boldsymbol{Q} = \bar{\boldsymbol{q}} + (h/2)\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \qquad \boldsymbol{P} = \bar{\boldsymbol{p}} - (h/2)\nabla U(\boldsymbol{Q}).$$

The composition simplifies to

$$\begin{aligned} \bar{\boldsymbol{p}} &= \boldsymbol{p} - \frac{h}{2}\nabla U(\boldsymbol{q}), \\ \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \\ \boldsymbol{P} &= \bar{\boldsymbol{p}} - \frac{h}{2}\nabla U(\boldsymbol{Q}). \end{aligned}$$